

T = 0.1 seg

ξ = 0.05

Wn = 62.831853

t* = 0.005

Wn = 2Π/T

Wa = 62.753264

$\omega_a = \omega_n \sqrt{1 - \xi^2}$

$u(t^*) = A + B t^* + C e^{-\xi \omega_n t^*} \cos \omega_d t^* + D e^{-\xi \omega_n t^*} \sin \omega_d t^*$

$\dot{u}(t^*) = B + (D \omega_d - C \omega_n \xi) e^{-\xi \omega_n t^*} \cos \omega_d t^* - (C \omega_d + D \omega_n \xi) e^{-\xi \omega_n t^*} \sin \omega_d t^*$

$\ddot{u}(t^*) = e^{-\xi \omega_n t^*} \left[(2 C \omega_d \omega_n \xi + D (\omega_n^2 \zeta^2 - \omega_d^2)) \sin \omega_d t^* - (C (\omega_d^2 - \omega_n^2 \zeta^2) + 2 D \omega_d \omega_n \xi) \cos \omega_d t^* \right]$

n	t	ag (gals)	α	A	B	C	D	u(t)	ú (cm/s)	ü (t) (cm/s²)
Po =		0	$\alpha = \frac{u_{g1} - u_{g0}}{t^*}$	$A = \frac{u_{g0}}{\omega_n^2} - \frac{2 \xi \alpha}{\omega_n^3}$	$B = \frac{\alpha}{\omega_n^2}$	$C = u_o - A$	$D = \frac{1}{\omega_d} [\dot{u}_o + C \omega_n \xi - B]$	0	0	0
1	0	-0.00649	-1.298000	0.000001	-0.000329	-0.000001	0.000005	0.000000	-0.000016	-0.006285
2	0.005	-0.03355	-5.412000	0.000001	-0.001371	-0.000001	0.000022	0.000000	-0.000112	-0.031688
3	0.01	-0.05503	-4.296000	-0.000007	-0.001088	0.000006	0.000016	-0.000001	-0.000313	-0.047849
4	0.015	0.04242	19.490000	-0.000022	0.004937	0.000020	-0.000083	-0.000003	-0.000290	0.056266
5	0.02	0.37344	66.204000	-0.000016	0.016770	0.000013	-0.000271	-0.000002	0.000809	0.377915
6	0.025	0.89455	104.222000	0.000053	0.026400	-0.000055	-0.000411	0.000008	0.003878	0.837269
7	0.03	1.39838	100.766000	0.000186	0.025524	-0.000178	-0.000354	0.000040	0.008980	1.185001
8	0.035	1.70742	61.808000	0.000329	0.015656	-0.000290	-0.000121	0.000100	0.015041	1.219161
9	0.04	1.86624	31.764000	0.000420	0.008046	-0.000320	0.000095	0.000189	0.020605	0.989316
10	0.045	2.13676	54.104000	0.000451	0.013705	-0.000262	0.000097	0.000304	0.025061	0.779407
11	0.05	2.80351	133.350000	0.000487	0.033778	-0.000184	-0.000148	0.000440	0.029257	0.884569
12	0.055	3.95763	230.824000	0.000617	0.058468	-0.000178	-0.000474	0.000599	0.034954	1.373177
13	0.06	5.40385	289.244000	0.000886	0.073266	-0.000287	-0.000625	0.000794	0.043459	1.997199
14	0.065	6.71389	262.008000	0.001263	0.066367	-0.000469	-0.000389	0.001037	0.054238	2.277593
15	0.07	7.38395	134.012000	0.001647	0.033946	-0.000609	0.000293	0.001335	0.064283	1.710114
16	0.075	7.03377	-70.036000	0.001899	-0.017740	-0.000564	0.001279	0.001671	0.068590	0.007449
17	0.08	5.57023	-292.708000	0.001900	-0.074144	-0.000229	0.002263	0.002002	0.061712	-2.721387
18	0.085	3.2419	-465.666000	0.001599	-0.117955	0.000403	0.002883	0.002263	0.039834	-5.940991
19	0.09	0.54238	-539.904000	0.001039	-0.136759	0.001224	0.002875	0.002375	0.002516	-8.847954
20	0.095	-1.98867	-506.210000	0.000341	-0.128224	0.002033	0.002185	0.002268	-0.046656	-10.649239

	A	B	C	D	E	F	G	H	I	J
12010										
12011		PGA max =	75.49619					Umax =	0.04067231	
12012		PGA min =	-58.57662					Umin =	-0.03203293	
12013								UmaxAbs =	0.0407	
12014								A = ωn²*UmaxAbs =	160.5678	
12015										
12016										
12017										

$\ddot{u}(t) = u(t) * \omega_n^2$